ARISTOTLE UNIVERSITY
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Intro and data structures

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https://practical-stats-med-r.netlify.app/

Basic concepts

Getting started with R and Rstudio: Objects

Everything in R is an **object**.

For example:

- a single number
- a variable
- output (e.g., a result from analysis, a figure)
- a data frame is an object that is itself a collection of objects etc.

Getting started with R and Rstudio: Functions

Most of the computations in R involves using functions.

A function essentially has a **name** and a list of **arguments** separated by a comma.

Example: `seq(from = 5, to = 8, by = 0.5)`

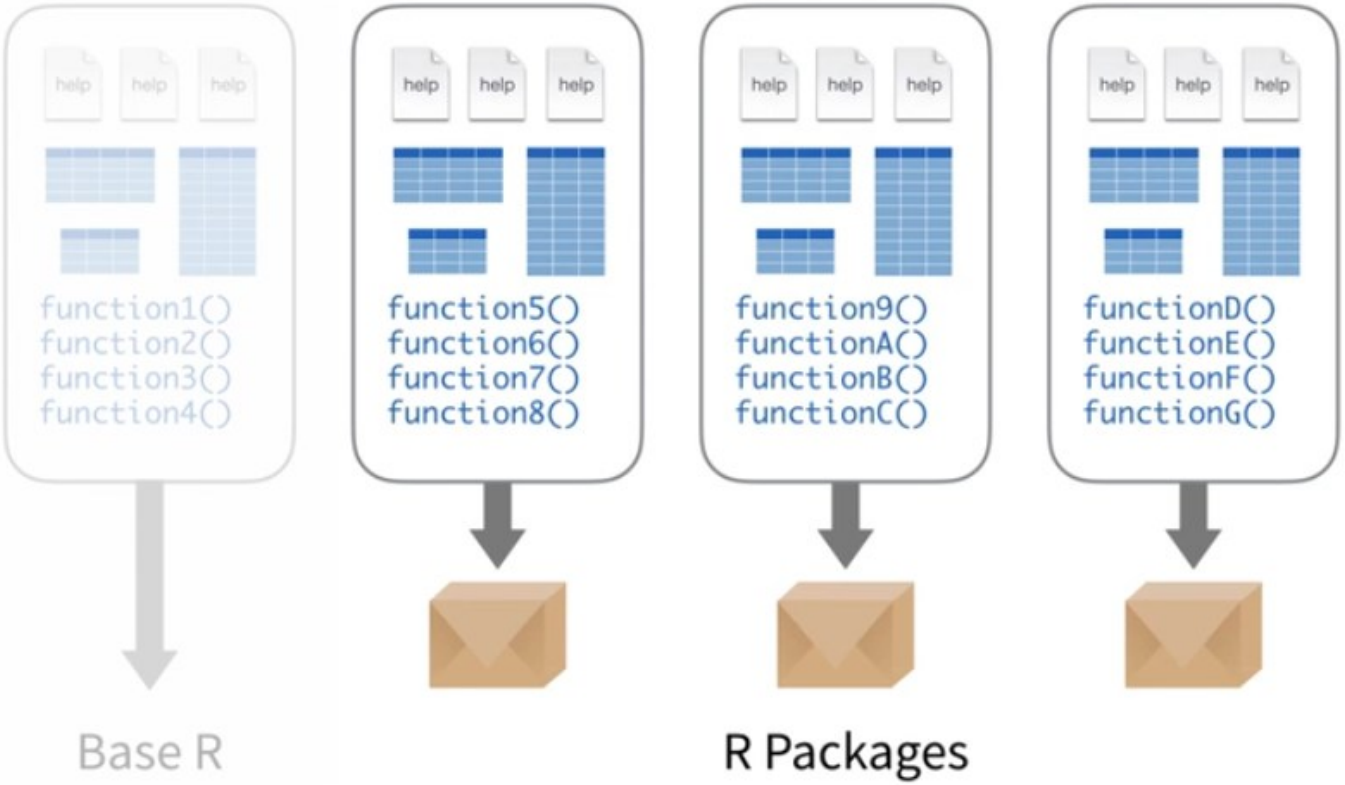
name

arguments

```
Output : [1] 5.0  5.5  6.0  6.5  7.0  7.5  8.0
```

Getting started with R and Rstudio: Packages

Functions "live" in Base R and in R packages!



Getting started with R and Rstudio: Operators

Arithmetic calculations with arithmetic operators

```
(3 + (5 * (2 ^ 2))) # hard to read
3 + 5 * 2 ^ 2       # clear, if we remember the rules
3 + 5 * (2 ^ 2)     # if we forget some rules, this might help
```

Relational (comparison) operators

<	>	==	<=	>=	!=

```
1 == 1  # equal to (double equal sign for equality)
```

Getting started with R and Rstudio: Math functions

Mathematical functions

```
sin(pi/2)  # trigonometry functions, angles in radians
```

```
log(100)  # natural logarithm
```

```
log10(100) # base-10 logarithm
```

```
exp(0.5) # e^(1/2)
```

Getting started with R and Rstudio: Scientific notation

Scientific notation is a special way of expressing numbers that are too big or too small to be conveniently written in decimal form.

- 0.000000015 is written 1.5×10^{-8}
  because 0.000000015 = 1.5 × 0.00000001 = 1.5 × 10^{-8} or 1.5e-8

- 150000000 is written 1.5×10^{8}
  because 150000000 = 1.5 × 100000000 = 1.5 × 10^{8} or 1.5e8

Getting started with R and Rstudio: Special values

Missing values (NA)

R uses a special numeric value NA standing for Not available. Mathematical operations using NA produces NA:

```
cos(NA)
```

```
## [1] NA
```

Not A Number (NaN)

The value NaN represents an undefined value

```
Inf/Inf
```

```
## [1] NaN
```

Infinitive: -Inf or Inf

```
1/0
```

```
## [1] Inf
```

Getting started with R and Rstudio: Named storage of objects

Assignment operator <-

(keyboard shortcut: **Alt** + - for Windows and **Option** + - for Mac)

```
x <- 1/40
```

```
sentence <- "the cat sat on the mat"
```

Getting started with R and Rstudio: Legal object names

Object names must start with a letter and **can** contain letters, numbers, underscores (_) and periods (.).

They **cannot** start with a number or underscore, nor contain spaces at all. Moreover, they can not contain “reserved" words (e.g., TRUE, FALSE, if etc).

Different people use different conventions for long object names, these include:

- periods.between.words
- underscores_between_words *(suggested)*
- camelCaseToSeparateWords

We must **be consistent**.

Note R is **case-sensitive** – it treats capital letters differently from lower-case letters.

Data structures in R

Basic data structures

The most fundamental concept in base R are the **vectors**.

vector

Atomic vectors

All elements of the same type:

- logical
- integer
- double
- character

Lists (generic vectors)

Can have different types of elements

NULL

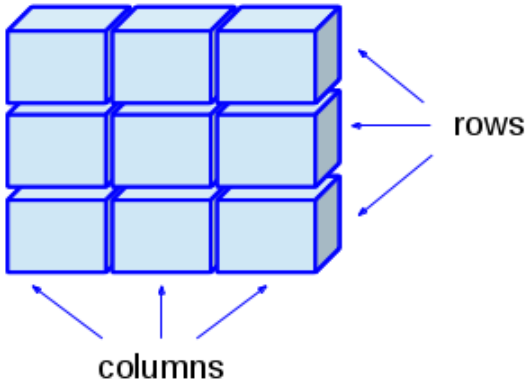
closely related to vectors and often serves the role of a generic **zero** length vector

Basic data structures

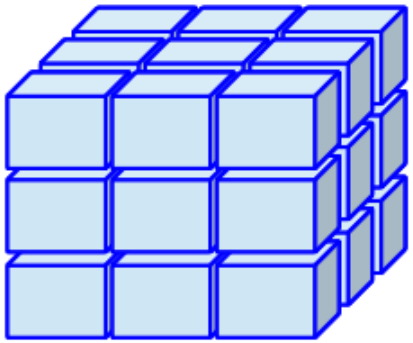
Atomic vectors

Vector

Matrix (**2D** vector)

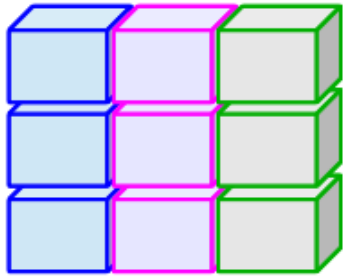


Array (**3D** vector)



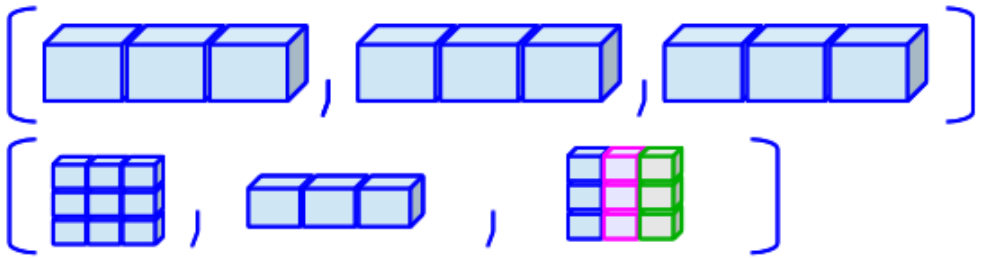
Lists (generic vectors)

Data Frame (Table)



Rectangular table

Lists



Basic data structures: vector

A collection of one or more objects of the same type. The combine function **c()** is used to form a vector. For example:

```
a <- c(4, 6, 20, 10) # numeric vector
```

```
b <- c("Male", "Female", "Male") # character vector
```

```
d <- c(TRUE, TRUE, TRUE, FALSE, TRUE) # logical vector
```

- Factor is a special vector that represents categorical data (eg. vector b). It can be ordered or unordered.

Basic data structures: matrix

A matrix is collection of data elements arranged in a two-dimensional rectangular layout. The following is an example of a matrix with 2 rows and 3 columns (2x3=6 elements):

```
a <- c(2, 1, 4, 5, 3, 7) # numeric vector with six elements

# Matrices are constructed column-wise starting in the "upper left" corner and running down the columns
A <- matrix(a, nrow=2, ncol=3)
```

1st column 2nd column 3rd column

1st row
2nd row

```
     [,1] [,2] [,3]
[1,]    2    4    3
[2,]    1    5    7
```

Basic data structures: data frame (a list with equal-length atomic vectors)

In a data frame the different columns can have **different types of data** (numeric, character, date etc.).

observations	age_yrs	sex	patient_is_male	adm_date	weight_kg	pain
	52	Male	TRUE	2009-10-15	78.3	mild
	57	Female	FALSE	2009-11-01	64.7	severe
	63	Female	FALSE	2009-10-21	67.2	moderate
	60	Male	TRUE	2009-10-28	NA	mild

```
df <- data.frame(age_yrs, sex, patient_is_male,
                 adm_date, weight_kg, pain)
```